CHEVALA SANDEEP

AI Agentic & Automation Engineer

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Summary

AI Agentic & Automation Engineer with hands-on experience designing, orchestrating, and shipping multi-agent automation systems using LangGraph, RAG pipelines, n8n, and AI coding agents like Claude Code. Built an end-to-end lead qualification and scheduling automation pipeline connecting CRM, WhatsApp, Calendar, Slack, and Email through agent-driven decision-making, plus a real-time ML-powered platform. Comfortable across the full stack — Python, FastAPI, PostgreSQL — with a focus on reliable, production-ready automation rather than one-off prompting.

Technical Skills

- **Automation & Integration** — n8n Workflow Automation, CRM & Pipeline Automation, REST APIs & Webhooks, LLM/OpenAI Integration, Conditional Logic & Routing, Error Handling & Debugging, Data Mapping & JSON, Process Analysis
- **GenAI & Agent Orchestration** — LangGraph (StateGraph, typed state), Multi-Agent Systems, RAG Pipelines (FAISS, Chroma), LLM-as-Judge Evaluation, Agent Tool-Calling, Retry/Fallback Design (Tenacity), Prompt/Context Engineering, Gemini API
- **ML Systems & Deployment** — FastAPI, Microservices Architecture, Docker, Model Deployment, Redis, PostgreSQL, RESTful API
- **Programming & CS Fundamentals** — Python, Java, Data Structures and Algorithms (100+ LeetCode problems solved in Java)
- **Tools & Platforms** — Git/GitHub, Django, MySQL, Supabase, Claude Code, Cursor, Replit, Antigravity ,Codex

Professional Experience

AI/ML Engineer Intern, *Synycs* 03/2026 – 04/2026

- Built an end-to-end Personal Expense Intelligence Platform for real-time financial transaction analysis.
- Designed and developed microservices architecture using FastAPI for transaction processing and ML inference.
- Implemented machine learning models (TF-IDF + Logistic Regression + MiniLM) for transaction classification with high accuracy.
- Built anomaly detection system using Z-score and Isolation Forest to detect unusual spending patterns.

Web Development Intern, *Beyond Career* 02/2025 – 05/2025

- Developed and maintained responsive web applications using modern front-end technologies, collaborating closely with back-end developers to integrate server-side logic.
- Optimized web applications for maximum speed and scalability.
- Participated in code reviews and contributed to team development practices

Projects

Lead Generation Multi-Agent System, *n8n, Python, FastAPI, LLM (agent orchestration), PostgreSQL, CRM, WhatsApp/Email APIs*

- Designed an end-to-end multi-agent pipeline triggered by new CRM leads: n8n orchestrates duplicate detection, then routes leads to a LangGraph-based qualification agent for scoring.
- Implemented a LangGraph-based lead qualification agent with typed state (StateGraph), a Chroma-backed RAG research node, deterministic scoring/enrichment tools, and an LLM-as-judge evaluation edge with confidence-threshold retries.
- Built a Supabase-backed duplicate-detection layer and PostgreSQL persistence for leads, scores, and outreach history, exposed via a FastAPI /process-lead endpoint, with a two-layer retry system — Tenacity-based exponential backoff for transient failures and a graph-level retry loop that re-runs the research-to-judge cycle on low-confidence output.
- Designed a human-in-the-loop approval gate before handoff — on owner approval, a WhatsApp Business API scheduling agent books calls via Calendar, Email, and Slack — deployed end-to-end on n8n Cloud, FastAPI (Render), Vercel, and Supabase.

AI Email Assistant, *n8n, Google Gemini API, Gmail, Sheets*

- Automated end-to-end email intake and structured AI classification (Gemini API) — detecting priority, generating summaries, and recommending actions for each incoming message.
- Built conditional routing and normalization logic in n8n to organize processed emails and log structured results (classification, priority, summary) to Google Sheets.

Personal Expense Intelligence Platform, *Python, FastAPI, Scikit-learn, MiniLM, React, Redis, PostgreSQL, Docker*

- Architected a real-time, event-driven microservices platform (FastAPI, React, Vite) using Apache Kafka for high-throughput transaction ingestion, with a document parsing service (pdfplumber, Tesseract OCR) to extract data from multi-format bank statements (PDF, CSV, XLSX).
- Engineered an active-learning NLP categorization engine (TF-IDF + Logistic Regression) that retrains and hot-swaps the model in memory after user corrections, achieving 92% classification accuracy, paired with an Isolation Forest anomaly detection pipeline (Z-score normalization) to flag irregular or duplicate transactions.
- Developed a real-time React dashboard (Tailwind CSS, Recharts) with a Generative AI financial coach (Gemini API), using Server-Sent Events (SSE) for live transaction/ML inference feeds and TTL in-memory caching (cachetools) to reduce latency and API cost.

Education

B.Tech in Artificial Intelligence and Data Science, *Bonam Venkata Chalamayya Engineering College* 2021 – 2025 | Odalarevu
CGPA: 8.52

Intermediate Education, *Ravi Teja Junior College* 2019 – 2021 | Allavaram
Percentage: 89

Certifications

- AWS Academy Machine Learning Foundations
- ServiceNow Certified System Administrator